Prajan Elango

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.33
Maharishi Vidya Mandir Chetpat Chennai	Class XII (CBSE)	2024	95.60%
Chinmaya Vidyalaya Anna Nagar West Chennai	Class X (CBSE)	2022	96.40%

PROFILE

Materials Science and Metallurgical Engineering undergraduate at IIT Hyderabad and hands-on experience in materials characterization, thermodynamic modeling, microscopy, and XRD analysis. Interested in materials research, semiconductor materials, advanced characterization, and manufacturing technologies.

COURSEWORK

· Physics of Solids · Semiconductor Devices · Materials Synthesis & Characterization · Thermodynamics of Materials · Phase Transformations · High Entropy Alloys · Transport Phenomena · Iron and Steel Making · Fundamentals of Extractive Metallurgy

PROJECTS

XRD-Based Crystal Structure & Lattice Parameter Analysis Report

- Analyzed XRD patterns of NdTiO (NTO) and BaTiO (BTO) to determine crystal structure, interplanar spacing, and lattice parameters using Bragg's law.
- Indexed diffraction peaks and identified monoclinic and tetragonal crystal systems through phase analysis and lattice calculations.
- Interpreted peak sharpness and diffraction behavior to evaluate crystallinity and structural symmetry.

Mechanical Deformation Analysis of Aluminium using XRD Report

- Investigated the effect of mechanical deformation on aluminium using XRD analysis of undeformed, hammered, and crushed samples.
- Evaluated lattice strain and crystallite distortion through peak broadening (FWHM) and diffraction peak comparison.
- Correlated deformation severity with microstructural defects, strain accumulation, and crystallinity reduction.

Zinc Nanoparticle Synthesis via Sol-Gel Processing Report

- Synthesized Zinc nanoparticles using the sol-gel route with controlled processing conditions.
- Performed XRD characterization to identify crystalline phases and assess structural properties.
- Analyzed diffraction data to evaluate crystallinity and validate successful nanoparticle synthesis.

Design and Fabrication of an Optical Microscope Report

- Designed and Fabricated an Optical Microscope (8* Magnification) with custom lens alignment and stage mechanism.

Design and Fabrication of a Portable Microspectrometer Report

- Designed and Fabricated a Microspectrometer using Diffraction Grating and Endoscopy camera got spectral analysis.

Slag Thermodynamics & Reduction Analysis using FactSage Report

- Performed thermodynamic analysis using FactSage to study Ellingham diagrams, oxide reduction behavior, and metallization trends in iron ore systems.

- Generated CaO-AlO-SiO ternary phase diagrams and interpreted tie-line behavior for slag system analysis.

- Evaluated hydrogen-based reduction of hematite across varying temperatures and analyzed equilibrium metallization behavior.

Iron Ore Pelletization Study Report

- Studied the effect of bentonite binder concentration on the drop strength of iron ore blue dust pellets.

- Conducted pellet preparation and mechanical drop-strength testing across varying binder compositions.

- Analyzed the relationship between binder percentage and pellet mechanical integrity.

Water Model Experiment on Blast Furnace Raceway Report

- Performed water model experiments to analyze jet penetration behavior in blast furnace raceway simulations using image-based measurements.

TECHNICAL SKILLS

Materials & Characterization:

· X-Ray Diffraction (XRD) ·Optical Microscopy · Spectroscopy · Materials Characterization ·Phase Diagram Analysis · Metallographic Sample Preparation

Materials Engineering:

· Semiconductor Materials · Solid-State Physics · Spectroscopy · Thermodynamics ·Phase Transformations · High Entropy Alloys

Software :

FactSage - Python (NumPy, Pandas, Matplotlib, SciPy) - Solid Edge - Fusion360 - Creality

POSITION OF RESPONSIBILITIES

Teaching Assistant - Mechanics of Solids Confirmation

- Assisted in evaluation, doubt resolution, and assessment activities for undergraduate coursework.

Corporate Partnerships & Growth Coordinator - Finance & Consulting Club Confirmation

- **Horizon** and secured 1 lakh in corporate funding through targeted partnership initiatives.

ACHIEVEMENTS

- Placed in the Top 1% (State) in National Standard Examination in Chemistry (NSEC) 2023. Certificate

- Received Academic Excellence Award from the Director, IIT Hyderabad, for achieving the highest departmental CGPA across Semesters 2 & 3 (9.6/10; overall: 9.33/10). Certificate